Fiscal Year:	FY 2010	**Task Last Updated:**	FY 07/27/2011
PI Name:	T'ien, James S. Ph.D.		
Project Title:	Combustion of Solid Fuel in Very Low Speed Oxygen Streams		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Surface combustion, shs, fire safety and smoldering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	jst2@case.edu	**Fax:**	FY 216 368-3007
PI Organization Type:	UNIVERSITY	**Phone:**	4408295283
Organization Name:	Case Western Reserve University		
PI Address 1:	Mechanical and Aerospace Engineering		
PI Address 2:	620 Glennan Hall, 10900 Euclid Avenue		
PI Web Page:			
City:	Cleveland	**State:**	OH
Zip Code:	44106-1712	**Congressional District:**	11
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	05/01/1998	**End Date:**	10/31/2009
No. of Post Docs:	1	**No. of PhD Degrees:**	4
No. of PhD Candidates:	4	**No. of Master' Degrees:**	2
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Sacksteder, Kurt R	**Contact Phone:**	216-433-2857
Contact Email:	Kurt.R.Sacksteder@grc.nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Extended to 10/31/2009 per PI (Ed., 7/27/2011)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Sacksteder, Kurt (NASA Glenn Research Center) Ferkul, Paul (National Center for Microgravity Research)		
Grant/Contract No.:	NNC04AA58A		
Performance Goal No.:	2.4		
Performance Goal Text:			
Task Description:	This task consists of ground-based experiments using low-gravity aircraft and continued development of a comprehensive predictive model of flame spread and extinction over solid fuels. The overall objectives of the research program are to understand the flame spread and extinction phenomena of solids in low-speed forced concurrent flows in microgravity and in upward buoyant flows in partial gravity. Specifically, we are interested in knowing how gravity affects the flammability limits and spread behaviour of solid materials and how to infer the combustion behaviour of these materials in low gravity from test results in normal gravity. The significance of the task is to understand the material flammability in reduced gravity and material screening tests for		

	low gravity applications; this can contribute to a safer space exploration mission.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This study on the basic mechanism of material flammability contributes to the fire safety of human both on earth and in space travel and exploration. The PI has served as a member of the National Academy of Engineering (NRC) committee to identify innovative research needs to foster fire safety in the US.
Task Progress:	This research program is concerned with the effect of low-speed flow on the spreading and extinction processes over solid fuels. Specifically, we are interested in the Solid Inflammability Boundary At Low-speed (SIBAL). The project has passed the Science Concept Review and the experiment was originally scheduled to be performed in the ISS Combustion Integrated Rack. However, with a shift of NASA prority, the original space experiment has been cancelled and the project became a ground-based investigation. Despise this change the overall objectives of the project remain the same. But, due to the lack of experimental platform in space, emphases have been shifted to modeling and normal gravity tests. The hypothesis (more accurately the prediction from the theoretical model) is that due to radiative loss from the flame, a quenching limit exists in low-speed flow. The complete flammability boundary consists of a low-speed quenching branch and a high-speed blowoff branch. The merging point of the two branches defines the fundamental flammability limit. The flammability boundary is expressed mainly as a function of oxygen percentage and flow velocity. The original objectives of the flight investigations as stated in the 2002 SIBAL SCR are: 1) Experimentally verify the model predictions by measuring the extinction limits in low-speed concurrent flow at several oxygen percentages, flow velocities and sample widths. 2) Verify the existence and quantitatively determine the fundamental low-oxygen limit. 3) Measure the steady flame spread rates and visible flame lengths. Measure the 2-D surface temperature distribution in the pre-heat and the pyrolysis zones. Measure flame (gas-phase) temperatures at selected locations. Measure radiation from the flame and the solid. 4) Compare the experimental data with model predictions. It is very clear that these original objectives are directly related to spacecraft fire safety. Two new objectives are added in this proposing period in order to answer an important question on the practical material flammability screening process and to address the possible different environment for fire safety in space exploration: 5) Evaluate the present NASA 6001 material flammability test #1 and to establish the relationship between its test results in normal gravity with material flammability characteristics in micro- and partial gravities. 6) Investigate material flammability in low pressure and high oxygen environment. Accomplishment 1. A summary of the design and the testing of a novel solid burning device for the original SIBAL project were given in the journal paper (J1). This includes related experimental and computational results. 2. Gas radiation effects and the influence of Fuel vapor Lewis number on flame spread and extinction were investigated theoretically in (J2) and (J4). 3. One-sided flame spread over a composite fabric was studied in detail in (J3) and cyclic flame spread over Nomex was discovered in (J6) and in Kleinhenz's Ph.D. thesis (P2). 4. Results of detailed measurement of radiation properties of thin cellulosic fuels were presented in Pettegrew's Ph.D. thesis (P1). 5. The behavior of flammability limit at low oxygen atmosphere was investigated theoretically for thick solid slabs in both normal and micro-gravities. It was discovered that a hysteresis phenomenon exists between the side and the wake stabilized flames (J5). 6. Pressure modeling as a simulation tool for estimating the rate of upward flame spread in partial gravity was proposed (J7). Other scale modeling issues were discussed in the book article (B1). 7. The mechanisms leading to pressure flammability limits were examined in (J8). 8. Development of three-dimensional steady concurrent flame spread model over solids with radiation in both forced and buoyant flows were advanced in (P3) and (P4). The latter includes detailed comparison with three sets of experimental data. References: J1--Hsu, S.-Y. and T'ien, J. S. "Pressure extinction limits of non-premixed flames " Combustion Theory and Modelling 13:885-900 , Dec-2009 J2--Kleinhenz, J., Feier, I. I., Hsu, S.-Y., T'ien, J. S., Ferkul, P. V. and Sacksteder, K. R "Pressure modeling of upward flame spread and burning rates over solids in partial gravity" Combust. Flame 154: 637-643 (2008). , Jul-2008 J3--Kleinhenz, J. E. and T'ien, J. S "Combustion of Nomex III Fabric in Potential Space Habitat Atmospheres: Cyclic Flame Spread Phenomenon" Combust. Sci. and Tech., 179: 2153-2169 (2007), Nov-2007 J4--Kumar, A. and T'ien, J.S ".: A computational Study of Low Oxygen Flammability Limit for Thick Solid Slabs" Combustion and Flame 146: 366-378 (2006), Mar-2006 J5--Tolejko, K., Feier, I. and T'ien. J. "Effects of Fuel Lewis Number on Flame Spread over Solids" Proceedings of the Combustion Institute 30/2: 2263-2270 (2005), Jun- 2005

	J6--Kleinhenz, J., Ferkul, P., Pettegrew, R., Sacksteder, K. and T'ien, J. "One-sided Flame Spread Phenomena of a Thermally Thin Composite Cotton/Fiberglass Fabric" Fire and Material, 29: 27-37 (2005), Jan-2005 J7--Kumar, A., Tolejko, K.D. and T'ien, J.S. "A Computational Study on Flame-Solid Radiative Interaction in Flame Spread over Thin Solid-Fuel" Journal of Heat Transfer 126:611-620 (2004), Aug-2004 J8--Ferkul, P.V., Kleinhenz, J., Shih, H.-Y., Pettegrew, R., Sacksteder, K. and T'ien, J. "Solid Combustion Experiments Using a Continuous Fuel Dispenser for Use in Microgravity and Associated Model Computations" Microgravity Science and Technology: XV/2: 3-12 (2004), Jul-2004 B1--Books/Book Chapters T'ien, J. S. "Some partial scaling considerations in microgravity combustion problems, p.281-292" in "Progress in Scale Modeling" Ed. Kozo Saito Springer, Jun-2008 P1--Pettegrew, R. D. "Radiative characteristics of a thin solid fuel at discrete levels of pyrolysis: angular. Spectral, and thermal dependencies" Ph.D. thesis, Case Western Reserve University, Jan-2006 P2--Pettegrew, R. D. "Radiative characteristics of a thin solid fuel at discrete levels of pyrolysis: angular. Spectral, and thermal dependencies" Ph.D. thesis, Case Western Reserve University, Jan-2006 P3--Kleinhenz, J. E. "Flammability and Flame Spread of Nomex® and Cellulose in Space Habitat Environments" Ph.D. thesis, Case Western Reserve University, May-2006 P4--Feier Jr., I. I. "A numerical Study of Three-dimensional Flame propagation over Thin Solids in Purely Forced Concurrent Flow Including Gas-Phase Radiation" Ph.D. thesis, Case Western Reserve University, Jan-2007 P5--Hsu, S.-Y. "Flame spread and extinction over solids in buoyant and forced concurrent flows: model computations and comparison with experiments" Ph.D. thesis, Case Western Reserve University, May-2009
Bibliography Type:	Description: (Last Updated: 09/19/2025)
Articles in Peer-reviewed Journals	Hsu S-Y, T'ien JS. "Pressure extinction limits of non-premixed flames." Combustion Theory and Modelling 2009;13(5):885-900. http://dx.doi.org/10.1080/13647830903225284 , Nov-2009
Articles in Peer-reviewed Journals	Kleinhenz J, Feier II, Hsu S-Y, T'ien JS, Ferkul PV, Sacksteder KR. "Pressure modeling of upward flame spread and burning rates over solids in partial gravity." Combustion and Flame, 2008 Sep;154(4):637-43. http://dx.doi.org/10.1016/j.combustflame.2008.05.023 , Sep-2008
Articles in Peer-reviewed Journals	Kleinhenz JE, T'ien JS. "Combustion of Nomex(R) III fabric in potential space habitat atmospheres: cyclic flame spread phenomenon." Combustion Science and Technology 2007;179(10):2153-69. http://dx.doi.org/10.1080/00102200701386172 , Oct-2007
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Books/Book Chapters	T'ien JS. "Some partial scaling considerations in microgravity combustion problems." in "Progress in Scale Modeling: summary of the first International Symposium on Scale Modeling (ISSM I in 1988) and selected papers from subsequent symposia (ISSM II in 1997 through ISSM V in 2006)." Ed. K. Saito. New York : Springer, 2008. p. 281-292., Jun-2008
Dissertations and Theses	Feier II Jr. "A numerical Study of Three-dimensional Flame propagation over Thin Solids in Purely Forced Concurrent Flow Including Gas-Phase Radiation." Ph.D. thesis, Case Western Reserve University, January 2007. , Jan-2007
Dissertations and Theses	Hsu S-Y. "Flame spread and extinction over solids in buoyant and forced concurrent flows: model computations and comparison with experiments." Ph.D. thesis, Case Western Reserve University, May 2009. , May-2009